SentinelScan — Software & System Requirements Specification

Platform Prerequisites, Runtime Dependencies, and Hardware Specifications | Author: Ankita Kala

1. System & Hardware Specifications

- **Operating System:** Linux (Ubuntu 22.04+ / Debian 12+), macOS 13+, or Windows 11 with WSL2
- **CPU Minimum:** 2 Cores (4 Cores recommended for parallel scan workers)
- **RAM Minimum:** 2 GB RAM (4 GB recommended for concurrent backend + frontend dev servers)
- **Disk Space:** 500 MB free storage for runtimes, logs, SQLite database, and report exports
- **Network:** Active outbound Internet connectivity for resolving and auditing public HTTPS targets

2. Required Runtime Environments

- **Python Runtime:** Python 3.11, 3.12, or 3.14 (with venv and pip package managers)
- **Node.js Runtime:** Node.js 20.x or 22.x LTS, npm 10.x
- **Docker Engine (Optional):** Docker 24.0+ and Docker Compose v2 for containerized deployment

3. Python Backend & Security Engine Dependencies

Package	Version Constraint	Functional Purpose in SentinelScan
fastapi	>=0.110.0	Asynchronous ASGI web application framework
uvicorn[standard]	>=0.28.0	High-performance ASGI production server with uvloop
pydantic	>=2.6.0	Schema validation, typing models, and payload sanitization
pydantic-settings	>=2.2.0	Defensive environment variable and config management
sqlalchemy	>=2.0.28	SQLAlchemy 2.0 ORM engine for scans and findings persistence
alembic	>=1.13.0	Database schema versioning and deterministic migrations
httpx	>=0.27.0	Asynchronous HTTP client with connection sandbox & 2MB limit
cryptography	>=42.0.0	X.509 certificate parsing, SAN extraction, and TLS trust checks
reportlab	>=4.1.0	Executive & technical vector PDF report generation engine
pillow	>=10.2.0	Image rasterization and badge rendering in PDF reports
pytest / pytest-asyncio	>=8.0.0	Automated unit and integration test runner (100% pass rate)
ruff	>=0.3.0	Static code analysis, security auditing, and AST linting

4. Frontend UI & Dashboard Dependencies

Package	Version Constraint	Role in SentinelScan UI
react / react-dom	^19.0.0	Modern reactive UI library with concurrent rendering
typescript	~5.8.0	Strict static typing across all 9 security workspaces
vite	^6.2.0	Next-generation lightning-fast frontend build bundler
tailwindcss	^4.0.0	Tailwind CSS v4 utility design system and dark theme tokens

lucide-react	^1.16.0	Defensive cybersecurity and telemetry icon set
recharts	^2.15.0	Interactive risk score charts, grade distributions, and dials
axios	^1.8.0	Promise-based HTTP client with global error interception

5. Installation & Verification Protocol

To install and verify all project requirements on a clean environment:

```
cd backend && python3 -m venv venv && source venv/bin/activate
pip install -r requirements.txt
pytest -v # Expected: 23 passed in <1.0s (100% pass rate)
cd ../frontend && npm install --legacy-peer-deps && npm run build
```

Repository: https://github.com/koyel2004/webscan | License: MIT